Settimana: 19

Materia: Matematica
Classe: 5C
Data: 24/02/26

Argomenti

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$f(x) = x + a + \frac{b}{x} + \frac{c}{x^2}$ $\qquad f(x) = x + 2 + \frac{12}{x} + \frac{8}{x^2} \rightsquigarrow f(4) = 4+2+3+\frac{1}{2} = 19/2$

Trova a,b,c in modo che: 1) $x=4$ pto critico $\rightsquigarrow f'(4)=0$ (1)
2) flesso in $B=(-2;-4)$ $\rightsquigarrow f''(-2)=0$ (2)
$\rightsquigarrow$ Passa per B (3)

$f(x) = x + a + bx^{-1} + cx^{-2}$

$f'(x) = 1 + 0 - bx^{-2} - 2cx^{-3}$

$f''(x) = 0 + 2bx^{-3} + 6cx^{-4}$

$$\rightsquigarrow \begin{cases} (1)\ 1 - \frac{b}{16} - \frac{\cancel{2}c}{\cancel{64}_{32}} = 0 \\ (2)\ \frac{2b}{-8} + \frac{\cancel{6}^{3}c}{\cancel{16}_{8}} = 0 \\ (3)\ -2 + a + \frac{b}{-2} + \frac{c}{4} = -4 \end{cases}$$

$$\begin{cases} (1)\ 32 - 2b - c = 0 \\ (2)\ -2b + 3c = 0 \\ (3)\ -8 + 4a - 2b + c = -16 \end{cases} \rightsquigarrow \begin{cases} 2b + c = 32 \\ 2b = 3c \\ 4a - 2b + c = -8 \end{cases}$$

$$\begin{cases} 4c = 32 \rightsquigarrow c = 8 \\ 2b = 24 \rightsquigarrow b = 12 \\ 4a = -8 + 24 - 8 \rightsquigarrow 4a = 8 \rightsquigarrow a = 2 \end{cases}$$

Detto A il pto con ascisse $x=4$ (che è critico) cos'è A?
e che tipo di flesso è B? ('Cioè ?)

$$f'(x) = 1 - \frac{12}{x^2} - \frac{16}{x^3} = \frac{x^3 - 12x - 16}{x^3} \geq 0$$

$x=4$ soluzione

Ruffini:

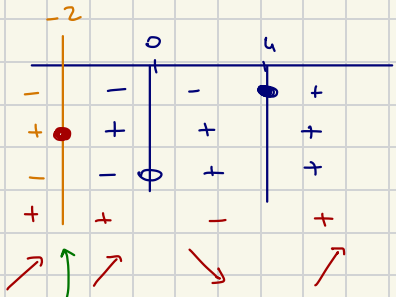
	1	0	-12	-16
4		4	16	16
	1	4	4	0

$(x^3-12x-16) = (x-4)(x^2+4x+4)$

N_1: $x-4 \geq 0 \rightsquigarrow x \geq 4$

N_2: $x^2+4x+4 \geq 0 \rightsquigarrow (x+2)^2 \geq 0 \quad \forall x \in \mathbb{R}$

D: $x^3 > 0 \rightsquigarrow x > 0$



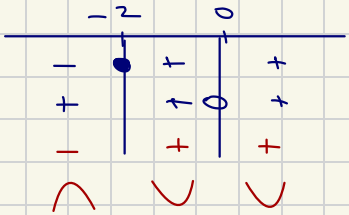
$\rightsquigarrow$ A è un minimo locale

Flesso a tg orizzontale

$$f''(x) = \frac{24}{x^3} + \frac{48}{x^4} = \frac{24(x+2)}{x^4} \geq 0$$

N: $x \geq -2$

D: $x^4 > 0 \rightsquigarrow x \neq 0$

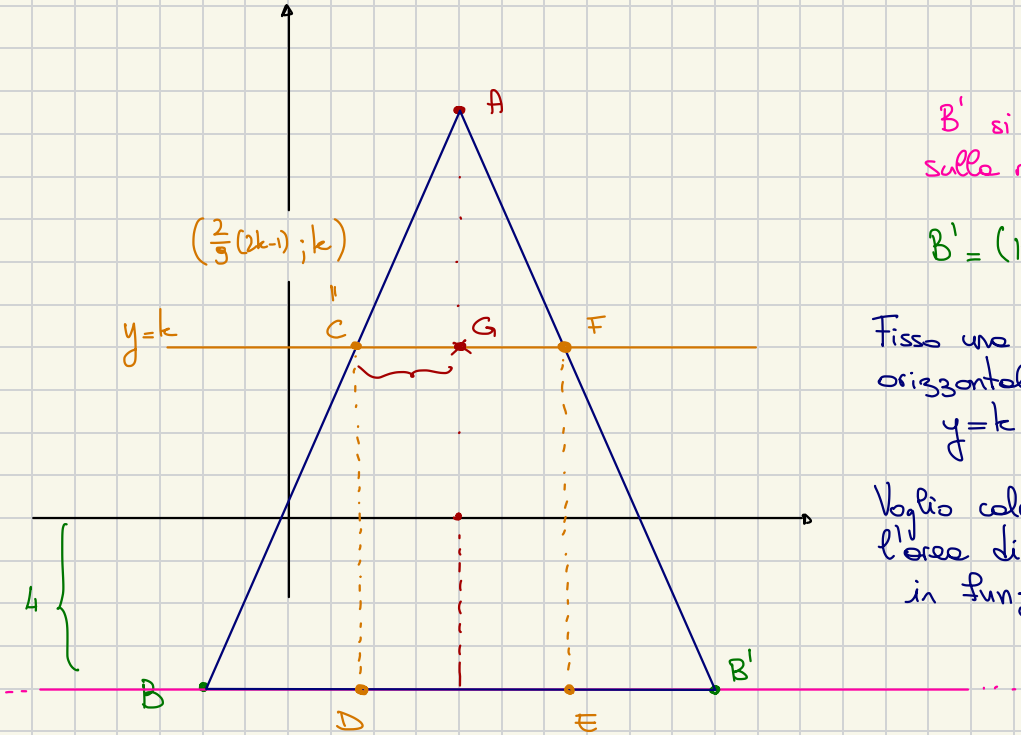


Il flesso è del tipo

Oss: Se per caso $x=-2$ è anche uno zero della derivata prima, allora il flesso è a tg orizz.

(c) Considera il triangolo ABB' Isoscele di base BB' parallela all'asse x e inscrivi in esso il rett. di area max

$A = \left(4; \frac{19}{2}\right)$, $B = (-2; -4)$



B' si trova sulla retta rosa

$B' = (10; -4)$

Fisso una retta orizzontale che è $y=k$

Voglio calcolare l'area di CDEF in funzione di k

$$h = CD = |y_C - y_D| = |k - (-4)| = |k+4|$$

Per trovare C, faccio $\cap$ tra retta AB e retta $y=k$

<u>Retta AB</u>: $\frac{y - y_B}{y_A - y_B} = \frac{x - x_B}{x_A - x_B} \rightsquigarrow \frac{y+4}{\frac{19}{2}+4} = \frac{x+2}{+4+2}$

$$\rightsquigarrow \begin{cases} 6(y+4) = \frac{27}{2}(x+2) \\ y = k \end{cases} \rightsquigarrow 6k + 24 = \frac{27}{2}x + 27$$

$$\rightsquigarrow x = \frac{2}{27}(6k-3) = \frac{2}{9}(2k-1)$$

$$CF = 2CG = 2|x_C - x_G| = 2\left|\frac{2}{9}(2k-1) - 4\right|$$

$$CF = 2\left|\frac{4k-2-36}{9}\right| = \frac{2}{9}|4k-38| = \frac{4}{9}|2k-19|$$

$$\underline{\text{Area}}(k) = CD \cdot CF = \frac{4}{9}|2k-19| \cdot |k+4|$$

$$f(k) = \frac{4}{9}|\underbrace{(2k-19)}_{k \geq \frac{19}{2}}\underbrace{(k+4)}_{k \geq -4}| = \frac{4}{9}|2k^2-11k-76|$$

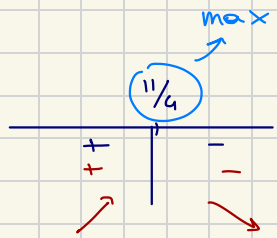
$$f(k) = \begin{cases} \frac{4}{9}(2k^2-11k-76) & k \leq -4 \ \vee \ k \geq \frac{19}{2} \\ -\frac{4}{9}(2k^2-11k-76) & -4 \leq k \leq \frac{19}{2} \end{cases}$$

↳ Per disegno ci interessa solo questo.

$$f: \left[-4; \frac{19}{2}\right] \longrightarrow \mathbb{R} \qquad f(k) = -\frac{4}{9}(2k^2-11k-76)$$

$$f'(k) = -\frac{4}{9}(4k-11) \geq 0$$

$$\cancel{\frac{4}{9}}(4k-11) \leq 0 \qquad k \leq \frac{11}{4}$$



Rettangolo di area max in corrispondenza della retta $y = \frac{11}{4}$

$$f\left(\frac{11}{4}\right) = -\frac{4}{9}\left(2\left(\frac{11}{4}\right)^2 - 11\left(\frac{11}{4}\right) - 76\right) =$$

Compito 3/3/26

Problema: 1) Parametri Lim o par
2) Studio di f_z (Rec fino a deriv. prima)
↳ Conclusione st. f_z.
3) Max min con un pto / Conseguenze Logs.

Q_1) 2 Lim + 2 Der	Q_5) Comb
Q_2) Limiti	Q_6) Tri
Q_3) Trova max min	Q_7) Nun con
Q_4) Conti – Der –	Q_8) Andis

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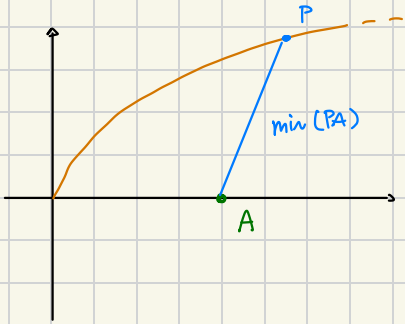
$f(x) = \sqrt{x}$ $\quad A = (4,0)$

Trova minimo di PA con P ∈ Curva.



$$P = (x; f(x)) = (x; \sqrt{x})$$

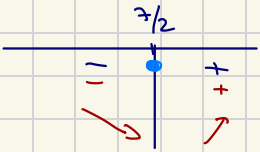
$$d(x) = \sqrt{(x_P - x_A)^2 + (y_P - y_A)^2} =$$

$$= \sqrt{(x-4)^2 + (\sqrt{x} - 0)^2} = \sqrt{x^2 - 8x + 16 + x}$$

$$d(x) = \sqrt{x^2 - 7x + 16}$$

Minimizzo $m(x) = x^2 - 7x + 16$

$$m'(x) = 2x - 7 \geq 0 \quad \rightsquigarrow \quad x \geq \frac{7}{2}$$



Il pto più vicino è $P = \left(\frac{7}{2}; \sqrt{\frac{7}{2}}\right)$

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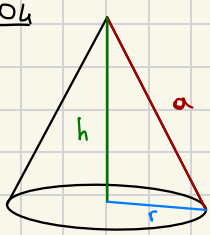


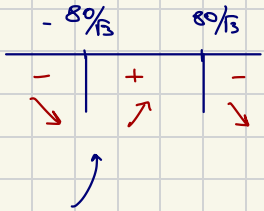
$a = 80$

Quanto vale il volume max del cono.

$$V = \frac{1}{3} S_{base} \cdot h$$

Scelgo h come variabile

$$r = \sqrt{80^2 - h^2} \quad \text{per teo di Pitagora}$$



$$V(h) = \frac{1}{3}\pi(80^2 - h^2) \cdot h = \frac{1}{3}\pi(80^2 h - h^3)$$

$$V'(h) = \frac{1}{3}\pi(80^2 - 3h^2) = \frac{1}{3}\pi(80 - \sqrt{3}h)(80 + \sqrt{3}h) \geq 0$$

$h = \frac{80}{\sqrt{3}}$ è il max $\rightsquigarrow$ $V(h_{max}) = \frac{1}{3}\pi \cdot \frac{2}{3}(80^2) \cdot \frac{80}{\sqrt{3}} = \frac{2}{9\sqrt{3}}\pi \cdot 80^3$

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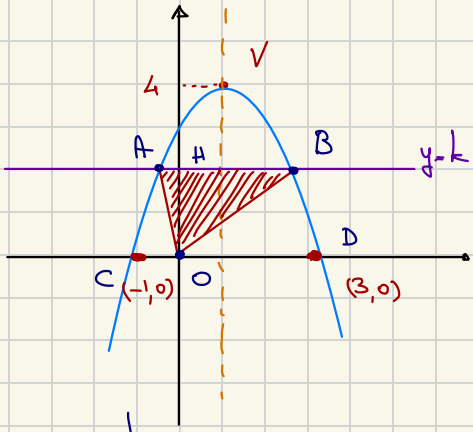
$y = ax^2 + bx + c \qquad V = \left(-\frac{b}{2a} ; -\frac{\Delta}{4a}\right)$



Il vertice ha coordinata x nel pto medio tra C e D

$\rightsquigarrow \quad V = (1 ; 4)$

$$\begin{cases} \frac{-b}{2a} = 1 \\ \frac{-b^2+4ac}{4a} = 4 \\ 0 = a - b + c \end{cases} \qquad \begin{cases} b = -2a \\ -\cancel{4}a^2 + \cancel{4}ac = \cancel{16}^{4}a \\ 3a + c = 0 \end{cases} \qquad \begin{cases} b = -2a \\ -a - 3a = 4 \\ c = -3a \end{cases}$$

$\rightsquigarrow \quad a = -1 , \quad b = 2 , \quad c = 3 \qquad \boxed{y = -x^2 + 2x + 3}$

$\triangle \quad OH = |k| = k \qquad 0 \le k \le 4$

$AB: \begin{cases} y = -x^2 + 2x + 3 \\ y = k \end{cases} \rightsquigarrow \quad x^2 - 2x + k - 3 = 0$

$\frac{\Delta}{4} = 1 - k + 3 = -k + 4$

$x_1 / x_2 = 1 \pm \sqrt{4-k} \qquad A = \left(1 - \sqrt{4-k} ; k\right)$

$B = \left(1 + \sqrt{4-k} ; k\right)$

$AB = \left| \cancel{1} + \sqrt{4-k} - \cancel{1} + \sqrt{4-k} \right| = 2\sqrt{4-k}$

$\text{Area}(k) = \frac{\cancel{2}k \cdot \sqrt{4-k}}{\cancel{2}} = \underset{k(4-k)^{1/2}}{k\sqrt{4-k}}$ Faccio la derivata ≥ 0

$$\text{Area}'(k) = 1\,(4-k)^{1/2} + k \cdot \frac{1}{2}(4-k)^{-1/2}(-1)$$

$$= \sqrt{4-k} - \frac{k}{2\sqrt{4-k}} = \frac{2(4-k) - k}{2\sqrt{4-k}}$$

$$= \frac{8-3k}{2\sqrt{4-k}} \geq 0$$

N: $8-3k \geq 0 \quad k \leq \frac{8}{3}$

D: $\forall k \in \text{Dominio}$

	$8/3$		4
+		−	−
+		+	+
+		−	−
↗		↘	↘

$k = \frac{8}{3}$ massimo

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$$f(x) = \frac{x^3}{6} - \frac{x^2}{2} - \frac{3}{2}x + \frac{11}{6} = \frac{1}{6}(x^3 - 3x^2 - 9x + 11)$$

(1) $\text{Dom}(f) = \mathbb{R}$

(2) Int: $x = 0 \quad y = \frac{11}{6} \qquad A = \left(0; \frac{11}{6}\right)$

$y = 0 \qquad x^3 - 3x^2 - 9x + 11 = 0 \qquad 1 - 3 - 9 + 11 = 0$

	1	−3	−9	11
1		1	−2	−11
	1	−2	−11	0

$(x-1)(x^2 - 2x - 11) \rightsquigarrow x = 1$

$\frac{\Delta}{4} = 1 + 11 = 12 \qquad x_1/x_2 = 1 \pm 2\sqrt{3}$

$B = (1-2\sqrt{3}; 0) \qquad C = (1; 0) \qquad D = (1+2\sqrt{3}; 0)$

(3) Segno: $\frac{1}{6}(x-1)(x^2-2x-11) \geq 0$

$N_1: \quad x \geq 1$

$N_2: \quad x \geq 1-2\sqrt{3}$

$N_3: \quad x \geq 1+2\sqrt{3}$

	$1-2\sqrt{3}$		1		$1+2\sqrt{3}$	
−		−		+		+
−		+		+		+
−		−		−		+
−		+		−		+

$f(x) \geq 0$ se $1-2\sqrt{3} \leq x \leq 1 \quad \vee \quad x \geq 1+2\sqrt{3}$

(4) <u>Limiti</u>: $\lim_{x \to +\infty} \frac{1}{6}\left(x^3 - 3x^2 - 9x + 11\right) = +\infty$

No Asintoti orizz
No Asintoti obl.

$\lim_{x \to -\infty} \frac{1}{6}\left(x^3 - 3x^2 - 9x + 11\right) = -\infty$

(5) <u>Derivata I</u>: $f'(x) = \frac{1}{6}\left(3x^2 - 6x - 9\right) = \frac{1}{2}\left(x^2 - 2x - 3\right)$

$f'(x) \geq 0 \qquad x^2 - 2x - 3 \geq 0 \qquad (x-3)(x+1) \geq 0$

		-1		3	
$x \geq 3$	$-$		$-$		$+$
$x \geq -1$	$-$		$+$		$+$
	$+$		$-$		$+$
	$\nearrow$		$\searrow$		$\nearrow$

$M = (-1; f(-1)) = \left(-1; \frac{8}{3}\right)$ Massimo

$m = (3; f(3)) = \left(3; -\frac{8}{3}\right)$ minimo

(6) <u>Derivata II</u>: $f''(x) = \frac{1}{2}(2x - 2) = x - 1 \geq 0$

		1	
$x \geq 1$	$-$		$+$
	$-$		$+$
	$\cap$		$\cup$

$F = (1; 0)$ Flesso

$M = \left(-1; \frac{8}{3}\right)$

$m = \left(3; -\frac{8}{3}\right)$

$C = (1; 0)$

$$\frac{x - x_m}{x_M - x_m} = \frac{y - y_m}{y_M - y_m}$$

$$\Downarrow$$

$$\frac{x-3}{-1-3} = \frac{y + \frac{8}{3}}{\frac{8}{3} + \frac{8}{3}}$$

$$(x-3)\frac{16}{3} = (-4)\left(y + \frac{8}{3}\right)$$

$$\frac{16}{3}x - 3 \cdot \frac{16}{3} = -4y - 2 \cdot \frac{16}{3}$$

$$\frac{16}{3}x + 4y = \frac{16}{3} \rightsquigarrow C$$

$$\rightsquigarrow \frac{16}{3} = \frac{16}{3} \quad \checkmark$$

(b) Traccia il grafico di $g(x) = |f(x)|$ e controlla la derivabilità.

- Disegno rosa

- Pti di <u>NON</u> deriv: Sono quelli dell'asse x. Cioè punti B, C, D. $\rightsquigarrow$ Uno si dovrebbe calcolare

$f'_-(x_B)$ $\quad$ $f'_+(x_B)$

$f'_-(x_C)$ $\quad$ $f'_+(x_C)$ $\quad$ Vengono opposti a coppie

$f'_-(x_D)$ $\quad$ $f'_+(x_D)$

<u>Pag 1912 n 83</u> $\quad f(x) = 3x\left(1 - \frac{x}{\sqrt{a+x^2}}\right) \quad a > 0$

(a) $f''(2) = 0 = f''(-2)$

$$f(x) = 3x - \frac{3x^2}{\sqrt{a+x^2}}$$

$$f'(x) = 3 - \frac{6x\sqrt{a+x^2} - 3x^2 \frac{1}{\cancel{2}} \frac{\cancel{2}x}{\sqrt{a+x^2}}}{a+x^2} = 3 - \frac{\overset{6ax + 6x^3 - 3x^3}{6x(a+x^2) - 3x^3}}{(a+x^2)\sqrt{a+x^2}}$$

$$= 3 - \frac{6ax+3x^3}{(a+x^2)^{3/2}}$$

$$f''(x) = -\frac{(6a+9x^2)(a+x^2)^{3/2} - (6ax+3x^3)\frac{3}{\cancel{2}}(a+x^2)^{1/2}\cdot\cancel{2}x}{(a+x^2)^3}$$

$$= \underbrace{-\frac{(a+x^2)^{1/2}}{(a+x^2)^3}}\,3\left[(2a+3x^2)(a+x^2) - (6ax+3x^3)x\right]$$

$$f''(2) = 0 \qquad \left[(2a+12)(a+4) - (12a+24)\cdot 2\right] = 0$$

$$2a^2 + 8a + 12a + \cancel{48} - 24a - \cancel{48} = 0$$

$$2a^2 - 4a = 0 \qquad 2a(a-2) = 0 \quad \begin{cases} a = 0 & \text{N.A.} \\ \boxed{a=2} \end{cases}$$

(b) Studio di funzione: $f(x) = 3x\left(1 - \frac{x}{\sqrt{2+x^2}}\right)$

(1) <u>Dom f</u>. Sempre poiché $2+x^2>0$ sempre vera $\mathbb{R}$

(2) <u>Assi</u>: $\underline{y=0} \rightsquigarrow 3x\left(1-\frac{x}{\sqrt{2+x^2}}\right) = 0 \rightsquigarrow x = 0$

$$1 - \frac{x}{\sqrt{2+x^2}} = 0 \rightsquigarrow \sqrt{2+x^2} = x \underset{x\geq 0}{\rightsquigarrow} 2+\cancel{x^2} = \cancel{x^2} \rightsquigarrow 2=0 \text{ imp}$$

$$\boxed{A = (0;0)}$$

$\underline{x=0}$

(3) <u>Segno</u>: $3x\left(1-\frac{x}{\sqrt{2+x^2}}\right) \geq 0 \qquad x \geq 0$

$$1 - \frac{x}{\sqrt{2+x^2}} \geq 0 \rightsquigarrow \frac{\sqrt{2+x^2} - x}{\sqrt{2+x^2}} \geq 0$$

$N \geq 0 \qquad \sqrt{2+x^2} - x \geq 0 \qquad \sqrt{2+x^2} \geq x \quad \rightsquigarrow \forall x \in \mathbb{R}$

$D \geq 0 \qquad \sqrt{2+x^2} > 0 \quad \rightsquigarrow \forall x \in \mathbb{R}$

$$\boxed{f(x) \geq 0 \quad \text{se} \quad x \geq 0}$$

(4) <u>Limiti</u> $\lim_{x \to +\infty} 3x \left(1 - \frac{x}{\sqrt{2+x^2}}\right) = \lim_{x \to +\infty} \frac{3x\sqrt{2+x^2} - 3x^2}{\sqrt{2+x^2}} = 0$

H) $\lim_{x \to +\infty} \frac{3\sqrt{2+x^2} + 3x \cdot \frac{1 \cdot \cancel{2}x}{\cancel{2}\sqrt{2+x^2}} - 6x}{\frac{1}{\cancel{2}} \frac{\cancel{2}x}{\sqrt{2+x^2}}} =$

$$\lim_{x \to +\infty} \frac{3(2+x^2) + 3x^2 - 6x\sqrt{2+x^2}}{x}$$

$$\lim_{x \to +\infty} \frac{6 + 6x^2 - 6x\sqrt{2+x^2}}{x} = \lim_{x \to +\infty} \frac{6}{x} + 6\left(x - \sqrt{2+x^2}\right) =$$

$$\lim_{x \to +\infty} \frac{6}{x} + 6\,\frac{\cancel{x^2} - 2 - \cancel{x^2}}{x + \sqrt{2+x^2}} = 0$$